Junyan Liu

jl7293@columbia.edu | linkedin | github | website

Education

Columbia University in the City of New York — New York, US
MS in Mechanical Engineering (Robotics and Control Track) | GPA: 3.92/4.33 — *Sep 2025 – Current*

University of Wollongong — Wollongong, AU
BE in Mechatronic Engineering (Honours) | WAM: 77.6/100 — *Jul 2024 – Jun 2025*

Beijing Jiaotong University — Beijing, CN
BE in Mechatronic Engineering | GPA: 3.19/4.00 — *Sep 2021 – Jun 2025*

Research Interests

World models and learning-based control for robotic manipulation; physical self-reproducing and self-reconfiguring modular robots; the intersection of mechanical design, control, and simulation.

Honors and Awards

Winner, Best Use of Computer Vision — MakeCU 2025 Hackathon, Columbia University — Nov 2025
- Selected from 19 hardware projects submitted by 115 registered participants at Columbia's flagship 24-hour hardware hackathon, hosted by the Columbia Robotics Club.

Research Experience

World Action Model for Robotic Manipulation — Jun 2026 – Present
RoboPIL Lab, Prof. Yunzhu Li (mentored by PhD student Kaifeng Zhang)
- **Objective:** understand when a learned world model actually helps a robot policy, by building World Action Models (WAMs) on the Interactive World Simulator (IWS), a latent video world model that predicts how a scene responds to robot actions; started from training-free MPPI baselines (Franka grasping in Newton, task planning in RoboCasa).
- On Push-T, benchmarked WAM variants built on frozen IWS dynamics against Diffusion Policy and a 450M SmolVLA; the best matched SmolVLA (171 vs. 172 of 250 episodes).
- **Key findings:** removing or scrambling IWS features breaks the policy, so it relies on the pretrained world model; reading IWS features beat imagining future frames first; and a VLA that trains its own vision gains little from IWS on Push-T, which motivated moving to harder tasks.
- Scaled to RoboDojo (Isaac Sim): built an automated data-collection pipeline (11.9M frames, 31k episodes) and trained IWS from scratch on this self-collected data.
- Reproduced the VPP2 baseline (14B video + 2B action) within leaderboard noise on three tasks; now training per-task WAMs (IWS → video DiT → action DiT) to compare against it.

Replicate Project: Self-Reproducing Modular Robots — Feb 2026 – Present
Creative Machines Lab, Prof. Hod Lipson
- **Objective:** test whether one homogeneous robotic cell can support distinct body plans and let a parent robot build an offspring from loose modules while keeping its own body intact ("robotic budding").
- Co-developed the control stack for a 15-module self-reproducing robot: ESP32 modules execute ESP-NOW-synchronized locomotion, module-carrying, and budding sequences across 5-module "seal" and 3-module "trilobite" morphologies.
- Built a closed-loop vision system (OpenCV, ArUco) that tracks module poses and steers the robot to dock, achieving fully autonomous collection of all scattered modules in physical trials.
- Engineered a custom magnetic physics engine for MuJoCo ($O(N^2)$ model with distance attenuation and damping) whose docking-force parameters were adopted for the physical docking interface.
- Migrated simulation to MuJoCo/MJX and wrote a C library injecting forces directly into MuJoCo's low-level arrays, achieving a 2.34× per-step speedup for large-scale evolutionary search.
- **Key findings:** the full budding sequence (collection, assembly, folding, separation, offspring locomotion) ran end to end for both morphologies, showing that one module design can realize different body plans through module count and connection topology alone.

Influence of Magnet Internal Structure on Magnetic Performance — Aug 2024 – Jun 2025
University of Wollongong, Dr. Chin-Hsing Kuo
- **Objective:** quantify how internal hollow structures change the magnetic force of cylindrical and rectangular NdFeB magnets.

- Built a dual-platform simulation: a custom MATLAB pipeline (Monte Carlo dipole-dipole modeling with 1,000,000+ sampling points per configuration, systematic geometry sweeps, cubic polynomial fitting) cross-validated against ANSYS Maxwell 2024 FEM (Virtual Work method).
- **Key findings:** magnetic force varies non-linearly (cubic) with hollow size; rectangular magnets start with higher force but decline more steeply than cylindrical ones, attributable to edge and corner flux distortion.

Legged Robot Learns to Walk via Reinforcement Learning Jan 2024 – Feb 2024

Global Education, Academics, and Research Skills (GEARS)

- Configured a Ubuntu-based RL simulation environment and designed reward functions to train a legged robot controller, evaluating walking performance across different reward compositions.

Publications

S. Zhang, **J. Liu**, W. He, and H. Lipson, "Robotic Budding: From Homogeneous Cells to Diverse Morphologies in Physical Self-Reproduction,"
Presented (poster) at the *IROS 2026 Workshop on Challenges and Applications Prospects for Reconfigurable Modular Robots* and the *IROS 2026 Workshop on Bridging the Gap between Robotic Matter and Active Metamaterials*, Pittsburgh, PA, USA, 2026.
arXiv:2610.31929, doi:10.48550/arXiv.2610.31929.

Teaching Experience

Columbia University New York, US

Teaching Assistant *Fall 2026*

MECE4058E: Mechatronics & Embedded Micro *Instructor: Enrico Zordan*

- Mentor 40 students in 7 teams as one of four TAs in weekly lab sessions and office hours (Tue/Thu, 2.5 h each); selected after completing the course with an A in Fall 2025.
- Guide students through debugging embedded microcontroller code (Assembly/C) and mechatronic hardware (solenoids, stepper/DC motors, thermal systems, magnetic levitation) across the course's case studies.
- Answer students' case-study questions outside lab by email and Canvas, and grade lab results.

Internship Experience

C-Based Structural Simulation Software Developer Nov 2024 – Feb 2025

Shenzhen Yunbo Software Technology Co., Ltd.

- Developed and optimized structural simulation software in C, enhancing performance of model setup, simulation core, and result visualization modules; improved simulation efficiency by 20%.
- Collaborated with a cross-functional team of 7 engineers to identify requirements and optimize algorithms, ensuring accuracy within 2% deviation from benchmark data.
- Conducted 50+ test/debug cycles, reducing runtime errors by 30% and increasing software stability.

Projects Design

Robotic Studio: Quadruped Robot "Pack-Pack" Development Sep 2025 – Dec 2025

- Designed and developed a custom quadruped robot from concept sketches to physical deployment, completing the entire pipeline of mechanical design, simulation, and control.
- Engineered the mechanical structure using SolidWorks for 3D CAD modeling and assembled the physical prototype integrating a Raspberry Pi core controller and 8 servo motors (2 DOF per leg).
- Constructed a high-fidelity physics simulation environment utilizing MuJoCo to safely and efficiently design, test, and evaluate various locomotion control strategies.
- Implemented and evaluated multiple control algorithms within the simulation, including preset gaits, Random Search, Hill Climbing, and Reinforcement Learning (RL), analyzing models optimized for maximum speed and gait stability.
- Successfully achieved sim-to-real transfer of heuristic search algorithms, deploying locomotion policies optimized via Random Search and Hill Climbing onto the physical hardware to achieve a stable real-world walking speed of 26.47 cm/s.

Google It: Computer-Vision Tracking Robot (MakeCU 2025 Hackathon) Nov 2025

- Built an autonomous mobile robot with a 4-person team in a 24-hour hackathon; pivoted from an initial two-wheel self-balancing design (PID control via MPU6050 IMU) to a three-wheel, camera-guided tracking robot after balance tuning proved infeasible within the time limit.
- Implemented real-time ArUco marker detection and tracking on a Raspberry Pi using OpenCV, computing proportional steering corrections from the marker's pixel offset to drive LX-16A bus-servo-actuated wheels toward the target.

- Integrated an HC-SR04 ultrasonic sensor for obstacle avoidance and an LED status indicator, halting the robot when an obstacle was detected or the target was lost.
- Diagnosed and fixed hardware failures under time pressure — burned-out motors, slipping wheels (resolved with paper shims and wire-wrapped tires), and insufficient battery current (upgraded from 9V to 12V with a DC-DC converter).

Lumbar Surgery Intelligent Planning System May 2023 – May 2024

- Developed an intelligent planning system for lumbar surgery based on X-ray images.
- Utilized deep learning algorithms (U-Net) for automatic lumbar parameter measurements and combined previous surgical data for intelligent surgery planning.
- Aimed to improve surgery efficiency and accuracy, reducing risks through intelligent decision-making support; collaborated with a team under the supervision of Prof. Tian Lixia.

Mechatronics Engineering Cup – Smart Car Challenge Mar 2023 – Jul 2023

- Built an STM32F103VCT-based smart car integrating infrared, ultrasonic, flame, photoresistor, and color sensors to achieve obstacle avoidance, line tracking, fire extinguishing, color-based sorting, and light-source-seeking behaviors.
- Implemented the embedded control stack: PWM motor control, ADC/DAC signal processing, SPI/I2C sensor interfacing, USART communication, and OLED display output.
- Designed individual wheel control logic and integrated it into unified car operation; tested and optimized across multiple terrains to validate reliable multi-function performance.

Ice Mark Grip Oct 2021 – Dec 2021

- Designed a procedure to command the robot's movement in two dimensions and search for objects using sensors.
- Applied hinges, sensors, and basic parts to allow motors to control planes of motion; the robot approaches with water after receiving a signal from the switch, and dispenses water after sensing the object's weight.
- Sensed instructions from the switch: the robot carries water toward the object from a distance and delivers it after serving.

Technical Skills

Programming & Embedded Systems: C/C++, Python, MATLAB, Arduino (ESP32), STM32
Robotics, Simulation & ML: MuJoCo/MJX, Deep Learning (U-Net)
Design & Analysis: SOLIDWORKS, AutoCAD, Inventor, Photoshop, ANSYS Workbench, ANSYS Maxwell
Tools & OS: Linux/Ubuntu, Git